Sampling a Distribution Randomly (Survey Sampling)

⇒ Large: Population $\{x_1, \dots, x_N\}$

⇒ Small: Sample $\{X_1, \dots, X_N\}$

Population Statistics:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \quad ; \quad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2 = \frac{1}{N}\left(\sum_{i=1}^{N} x_i^2 - \mu^2\right)$$

Sample Statistics:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad ; \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

→ $\binom{N}{n}$ possible random samples each equally likely

This process is called "simple random sampling without replacement.

For any individual sample X_i:

$E(X_i) = \mu \qquad \mathrm{Var}(X_i) = \sigma^2$

$$\sum_{j=1}^{N} x_j \cdot P(X_i = x_j) = \sum_{j=1}^{N} x_j \cdot \frac{1}{N} = \mu$$

Expected Value of the Sample Mean

$E(\bar{X}) = \mu$ ($\bar{X}$ is an unbiased estimate of μ)

$$E(\bar{X}) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) =$$

$$= \frac{1}{n}\sum_{i=1}^{n}\mu = \mu$$

Variance Value of the Sample Mean

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n}(X_1 + \dots + X_n)\right) =$$

$$= \frac{1}{n^2}\left[\mathrm{Var}(X_1) + \dots + \mathrm{Var}(X_n)\right] =$$

if X_i's are indep. & $N \gg 1$

$$= \frac{\sigma^2}{n}$$

← CLT result

For finite population size N

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}\left[1 - \frac{n-1}{N-1}\right] \approx \frac{\sigma^2}{n} \quad \text{for } n \ll N$$